

A Real-time Visualization System of Pseudo-muscle Activity Using a VR Device

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ABSTRACT

The goal of this research is to provide a real-time display of muscle activity in response to human motion input. We developed a system for real-time visualization of pseudo-muscle activity using a VR device. The user controls a CG avatar in VR space by inputting a body motion with the VR device. We constructed simplified models to output muscle activity corresponding to the posture of a CG avatar based on muscle-analysis data estimated from motion data. This system allows users to visualize the position of active muscles using a CG avatar in a VR space. This system can present different muscle activities depending on the input of elbow flexion/extension movement and the state of the upper limb at that time. The position of active muscles and the magnitude of their activity are represented by the color and expansion of the muscles on a CG avatar. By assigning controller input to other parts of the body rather than just the hands, users can move those parts in virtual space and visualize muscle activity.

CCS CONCEPTS

• **Computing methodologies** → **Virtual reality**; • **Human-centered computing** → *Visualization systems and tools*.

KEYWORDS

Muscle activity, Visualization, Virtual reality

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1 INTRODUCTION

In recent years, experiential content using input from body movements with VR (Virtual Reality) has emerged, including studies that visualize muscle activity during exercise. However, calculations of muscle analysis are time consuming and feedback of muscle activity in real-time for users is difficult in general. The purpose of this study is to recognize the muscles used in the user's movements

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Figure 1: Visualization system of muscle activity.

to support exercise in sports, rehabilitation, and other physical activities. We have developed a system that visualizes the position and pseudo-muscle activity used during exercise in the VR space.

Choi et al. [Choi et al. 2007] used an optical motion capture system and an EMG (Electromyography) device to analyze quantitative movement and to visualize muscle activity during movement. Some research use AR (Augmented Reality) and 3D animation for visualization by highlighting the joint positions of patients and athletes [Zhou et al. 2023][Hurter and McDuff 2017]. Our study aims to develop a system that provides real-time feedback of muscle activity during body movements. In the proposed system, each part of the body is manipulated through the controller's input in VR, without using full-body input. The 3D avatar can be operated from the controller's input with VR to visualize muscles during body movements, and thus the user can view muscle activities that are otherwise difficult to view. Therefore, our system is expected to provide the user with a deeper understanding of full-body muscle activity during exercise.

2 VISUALIZATION SYSTEM USING A VR DEVICE

Figure 1 shows a scene of our system running. This system controls a 3D avatar's body in real-time based on inputs of positions and directions from a head-mounted display (HMD: Meta Quest 2) and controllers held in both of the user's hands. The movement of the arms that manipulate the 3D model during operation is controlled by IK (Inverse Kinematics). Magnitude of muscle activity corresponding to the input movement is modeled in advance and visualized in real time using color shading of CG avatar's muscle. The

model is based on the muscle-analysis data estimated from motion data, corresponding to the information input from the controller, and 12 muscles are visualized in the upper and lower extremities. The user views the 3D avatar in VR space from the side with a mirror or from behind using a third-party view. This system was developed using Unity and the programming language used was C#.

This paper describes a simplified model to output muscle activity corresponding to the posture of a CG avatar based on muscle-analysis data estimated from motion data. This paper also describes a visualization method and results that expresses the magnitude of muscle activity through color shading the location of the CG avatar's active muscles.

3 VISUALIZATION METHODS AND RESULTS

We created a prototype model that visualizes muscle activity in real time based on estimated muscle activity from pre-captured motion data, as well as basic biomechanics theory. Based on the motion data and muscle activity estimation of the actor, we found that when the elbow is slowly bent vertically while the upper arm is kept horizontal, the strength of the biceps muscle peaks when the elbow is in a flexed position of approximately 45 degrees. Theoretically, further flexion reduces the use of the biceps and uses the triceps instead. We found that when maximally flexing the elbow, the triceps are largely used, and the biceps are hardly used.

In this system, we created a simple mathematical model of the quadratic curve based on this result. The elbow angle of the CG avatar that is synchronized with the user's movements is used as input. Regarding the output, the amount of muscle activity is displayed in shades of color, and the color becomes red when the muscle is fully used.

Figure 2 shows the visualization results of the system. When bending the elbow to about 45 degrees, the CG model of the biceps turns red. As you bend your elbow, you will see a gradual increase in the amount of muscle activity in your triceps, while maintaining the total balance of biceps and triceps activity. Users can visualize the elbow angle at which muscle activity peaks while moving their arm.

The advantage of this system is that it can provide visual feedback in real time according to human body movements input by the user. By assigning controller input to not only the hand but also other body parts and moving them, it is also possible to visualize the muscle activity of that body part. Figure 3 is an example of controlling the legs of a CG avatar using a VR controller. Regarding the hip joint, you can see the positions of the muscles and their amount of activity used in adduction/abduction (Fig. 3(a)), external rotation/internal rotation (Fig. 3(b)), and their combinations (Fig. 3(c)). Muscle activity that is difficult to see in the real world can be previewed in a more comfortable posture with a mirror or in third-person perspective in VR space.

4 CONCLUSION

We developed a system for real-time visualization of pseudo-muscle activity using a VR device. Experiments were conducted with 30 subjects to assess the effectiveness of the system's concept and to explore its visualization methods. As a result, the system was



Figure 2: Visualization results in bending elbow.

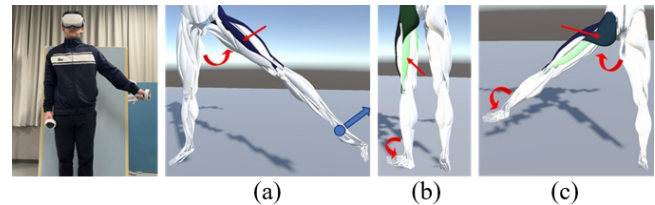


Figure 3: Visualization of muscle activity in the lower extremity by hand controller input.

highly evaluated as a means of understanding and improving body motion from subject feedback, thus confirming the usefulness of this system. The created model is based on muscle analysis data estimated from a single actor, and does not take into account the person's body shape or muscle mass. By collecting and utilizing a large amount of motion data, it becomes possible to respond to user variations. In future works, we will incorporate a prediction model in the system by modeling muscle analysis data from motion data as a way to analyze muscle loading.

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